

FRAUDULENT CLAIM DETECTION

Report Type	Predictive Analytics — Fraud Detection
Dataset	insurance_claims.csv (1,000 records, 40 features)
Models Evaluated	Logistic Regression Random Forest (Tuned)
Team	Vaibhav Fulara, Tarun Garg, Banetta Nedunchezhiyan, Sharath Hosakote, Karthik Reddy

1. EXECUTIVE SUMMARY

Global Insure faces a significant and growing threat from fraudulent insurance claims. This report presents findings from a comprehensive data-driven analysis of 1,000 historical claims across 40 attributes, aimed at building an early-warning fraud detection system that can flag suspicious claims at the point of submission — before payouts are made.

The analysis followed a rigorous end-to-end machine learning pipeline: data preparation, exploratory data analysis (EDA), feature engineering, model building, and validation. Two models were evaluated — Logistic Regression and a hyperparameter-tuned Random Forest. The tuned Random Forest emerged as the recommended production model, achieving 84.67% accuracy and an AUC-ROC of 0.893 on the held-out validation set.

Key Outcome: The tuned Random Forest model correctly identifies 81% of fraudulent claims on unseen data (Sensitivity = 81.08%), while maintaining a Specificity of 85.84% — meaning only 1 in 6 legitimate claims is flagged for unnecessary review. Deployment of this model is estimated to prevent significant financial losses annually.

Top Findings at a Glance

- Hobby is the single strongest fraud predictor: chess & cross-fit policyholders show ~80% fraud rate.
- Major Damage incidents are MORE fraudulent than Minor or Total Loss.
- High vehicles claim relative to injury claim is a statistically significant red flag.
- Ohio (OH) and Pennsylvania (PA) account for disproportionate fraud volumes.
- Documentation gaps signal fraud: no police report combined with property damage and multiple witnesses is a red flag combination.
- Both models substantially outperform random guessing (AUC 0.877–0.893 vs baseline 0.500).

2. PROBLEM STATEMENT & OBJECTIVES

Insurance fraud is a global epidemic that inflates premiums for honest policyholders and erodes profitability for insurers. At Global Insure, a meaningful percentage of claims processed each year are fraudulent, and these are typically detected only after full or partial payouts — at considerable financial and operational cost.

Business Objectives

Primary Goal	Build a predictive model to classify claims as fraudulent or legitimate at intake, before payment is released.
Secondary Goal	Identify the most predictive features of fraud to inform manual review prioritisation and policy design.
Analytical Goal	Uncover patterns in historical claims data that reveal fraud risk segments and geographic hotspots.
Strategic Goal	Provide actionable recommendations to reduce fraud exposure across underwriting, claims, and operations.

Key Questions Answered

- Which policyholder characteristics and claim attributes most strongly predict fraud?
- Are there demographic, occupational, or geographic segments with elevated fraud risk?
- Can fraud be detected early enough (at intake) to prevent payout?
- Which machine learning model offers the best trade-off between fraud detection and false alarm rates?

3. DATASET OVERVIEW

The analysis was conducted on the insurance_claims.csv dataset provided by Global Insure, comprising 1,000 claim records and 40 features spanning policyholder demographics, vehicle information, incident details, and claim financials.

Dataset Dimensions

Total Records	1,000 claims
Total Features	40 columns (21 categorical, 17 integer, 2 float)
Target Variable	fraud_reported (Y = Fraudulent, N = Legitimate)
Class Distribution	~25% Fraudulent (250) · ~75% Legitimate (750)
Date Range	Policies from mid-1990s to 2015; incidents spanning multiple years

Feature Categories

Policy Attributes	policy_state, policy_csl, policy_deductable, policy_annual_premium, umbrella_limit
Policyholder Demographics	age, months_as_customer, insured_sex, insured_education_level, insured_occupation, insured_hobbies
Incident Details	incident_type, collision_type, incident_severity, incident_state, incident_city, incident_hour_of_the_day
Claim Financials	total_claim_amount, vehicle_claim, injury_claim, property_claim
Supporting Evidence	authorities_contacted, police_report_available, property_damage, number_of_vehicles_involved, witnesses, bodily_injuries
Vehicle Details	auto_make, auto_model, auto_year

4. DATA PREPARATION & CLEANING

4.1 Handling Null and Invalid Values

Several columns contained the placeholder '?' representing missing data. These were replaced with NaN and handled using different imputation strategies:

collision_type (178 nulls)	Filled with mode ('Rear Collision') — most common collision type
authorities_contacted (91 nulls)	Logically inferred from police_report_available; remainder filled as 'Other'
property_damage (360 nulls)	Derived from property_claim: 0 → 'NO', >0 → 'YES'
_c39 (1,000 nulls)	Completely empty column — dropped entirely

4.2 Column Removal

Four columns were removed for being identifiers, near-identifiers, or fully empty:

- `_c39` — 100% null, no information value
- `policy_number` — unique identifier per row, no predictive power
- `incident_location`, `insured_zip` — 99%+ unique values, hence very less predictive power or potential for pattern identification

4.3 Logical/Illogical Value Removal

- Rows with `umbrella_limit < 0` were removed (negative insurance limits are meaningless).
- Rows where `policy_annual_premium ≥ total_claim_amount` were removed (illogical financial relationship).

4.4 Data Type Corrections

- `incident_date` converted from object to `datetime64`.
- `policy_bind_date` converted from object to `datetime64`.

4.5 Final Dataset After Cleaning

Records	998 rows (2 removed as illogical)
Features	36 columns (4 dropped)
Nulls Remaining	0 — all handled

5. TRAIN-VALIDATION SPLIT & CLASS BALANCE

To ensure unbiased model evaluation, the cleaned dataset was split into a 70% training set and a 30% validation set using stratified sampling on the target variable.

Training Set	698 rows (stratified)
Validation Set	299 rows (stratified)
Stratification	On <code>fraud_reported</code> to preserve class proportions in both sets
Random State	42 (reproducible split)

Handling Class Imbalance

The training set exhibited a ~75:25 imbalance (Legitimate:Fraud). To prevent the model from learning a bias towards the majority class, `RandomOverSampler` was applied exclusively to the training data:

Original Training	698 rows (~525 Legit, ~173 Fraud)
After Oversampling	1,050 rows (balanced)
Method	<code>RandomOverSampler</code> — duplicates minority class samples with replacement
Validation Set	Untouched — preserves real-world imbalance for honest evaluation

Important: Oversampling was applied only to training data. The validation set retains the original class distribution to ensure performance metrics reflect real-world conditions.

6. EXPLORATORY DATA ANALYSIS

6.1 Distribution of Numerical Features

Numerical features were analysed for distributional shape, skewness, and outliers. Key observations:

- `umbrella_limit`: Highly right-skewed — the majority of policyholders do not opt for umbrella coverage. Log transformation did not improve predictive value; feature was dropped.
- `total_claim_amount` & `vehicle_claim`: Bimodal distribution — one cluster of low-value trivial claims, and a second broader cluster of genuine mid-to-high value claims.
- `incident_hour_of_the_day`: Fraud incidents peak at hour 13 (afternoon) vs hour 11 for legitimate incidents.
- `capital_loss`: Over 50% of fraud cases show `capital_loss` = 0, suggesting fraudsters tend not to be financially distressed investors.

6.2 Correlation Analysis

A lower-triangle correlation heatmap was computed for all numerical features. Key findings:

- Strong collinearity cluster: `total_claim_amount`, `vehicle_claim`, `injury_claim`, and `property_claim` are highly inter-correlated ($r > 0.7$). These were consolidated into a single engineered feature (`claim_to_premium_ratio`).
- `age` and `months_as_customer` are highly correlated ($r \approx 0.88$) — `age` was dropped to avoid redundancy.
- `fraud_bin` shows highest correlation with `vehicle_claim` (positive) and `incident_severity`-derived features.

6.3 Class Balance Check

The training set showed ~25% fraudulent claims. After `RandomOverSampler`, the training set was balanced to 50/50, while the validation set retained the original ~25% fraud proportion.

6.4 Bivariate Analysis — Categorical Features

6.4.1 Fraud Rate by Hobby

Hobby is the most discriminating categorical feature. Policyholders reporting chess or cross-fit as hobbies show a fraud rate of approximately 80% — nearly 3× the dataset average.

chess	~80% fraud rate — highest risk hobby
cross-fit	~78% fraud rate — second highest
Average	~25% across all hobbies
Implication	Incorporate hobby risk-scoring at policy issuance stage

6.4.2 Fraud Rate by Occupation

- `exec-managerial`: Elevated fraud rate — highest among all occupations.
- `transport-moving` combined with high education (MD/Masters/PhD): ~45% fraud rate.
- Highly educated policyholders (Masters, PhD, MD) collectively over-represent fraud cases, suggesting more sophisticated scheme execution.

6.4.3 Incident Severity — The Severity Paradox

Incidents classified as 'Major Damage' have the highest fraud rate, while 'Minor Damage', 'Trivial Damage', and 'Total Loss' incidents are less fraudulent. This suggests that fraudsters deliberately stage moderate-severity incidents — serious enough to justify a large claim, but not extreme enough to trigger heightened scrutiny.

Major Damage	Highest fraud % — staged moderate incidents
Total Loss	Lower fraud % — harder to fake total destruction
Minor Damage	Lower fraud % — too small a payout to justify risk
Trivial Damage	Lowest fraud % — minimal claim value

6.4.4 Geographic Distribution

- Ohio (OH) and Pennsylvania (PA): Statistically elevated fraud rates above 30%.
- West Virginia (WV): Below-average fraud rate — lower-risk region.
- Fraud concentration suggests organised regional fraud networks rather than individual opportunism.

6.4.5 Documentation Patterns

- Police report NOT available: Strongly associated with fraud — fraudsters avoid official documentation.
- Property damage = YES without a police report: High-risk combination.
- ≥ 2 witnesses present AND no police report: Pre-arranged witnesses — organised staging pattern.

6.5 Bivariate Analysis — Numerical Features

6.5.1 Claim Amount Patterns

Fraudulent claims show a tighter, higher cluster in vehicle_claim and total_claim_amount distributions compared to legitimate claims. The primary driver of elevated total claims in fraud cases is the vehicle component — not injury or property.

- vehicle_claim: Fraud cases cluster at higher values with lower variance.
- injury_claim: Much weaker separation between fraud and legitimate — fraudsters under-report injuries.
- The vehicle-to-injury ratio (vehicle_claim / injury_claim) is a powerful engineered fraud signal.

6.5.2 Incident Hour

Fraud incidents have a slightly later median hour (13:00) compared to legitimate incidents (11:00). Afternoon incidents warrant additional scrutiny, particularly in high-risk states.

7. FEATURE ENGINEERING

Feature engineering transformed raw attributes into higher-signal predictors, capturing temporal, financial, and domain-specific fraud patterns.

7.1 New Features Created

claim_to_premium_ratio	$\text{total_claim_amount} \div \text{policy_annual_premium}$ — relative claim burden
vehicle_to_injury_ratio	$\text{vehicle_claim} \div (\text{injury_claim} + 1)$ — staged accident signal
net_capital_gain	$\text{capital_gains} + \text{capital_loss}$ — financial profile summary
vehicle_age	$\text{incident_year} - \text{auto_year}$ — older vehicles may indicate fraud
days_between_policy_and_incident	Days from policy binding to incident — early-claim flag input
early_claim_flag	1 if claim filed within 365 days of policy binding
is_high_risk_hobby	1 if hobby in {chess, cross-fit}
is_high_risk_occupation	1 if occupation = exec-managerial
highly_educated	1 if education in {Masters, PhD, MD}
is_high_risk_state	1 if incident_state in {OH, PA}
is_high_risk_auto_make	1 if auto_make in {Ford, Mercedes}
incident_hour_bucket	Hour categorised into 4 bands: 0-5, 6-11, 12-17, 18-23
policy_bind_month	Month extracted from policy_bind_date — seasonality signal

7.2 Features Dropped (Post Engineering)

- total_claim_amount — sum of components, replaced by claim_to_premium_ratio.
- age — highly correlated with months_as_customer.
- umbrella_limit — minimal predictive value even after log transformation.
- capital-gains, capital-loss — consolidated into net_capital_gain.
- insured_education_level — captured by the highly_educated binary flag.
- incident_hour_of_the_day — replaced by incident_hour_bucket.
- auto_year — replaced by vehicle_age.

7.3 Categorical Consolidation

Low-frequency categories were collapsed to 'Other' to reduce noise and prevent overfitting on sparse dummy variables. Thresholds were calibrated on training data and applied consistently to validation data.

insured_relationship	< 10% frequency → 'Other'
insured_occupation	< 10% frequency → 'Other'
insured_hobbies	< 10% frequency → 'Other'
incident_city	< 10% frequency → 'Other'
auto_make / model	< 10% frequency → 'Other'

7.4 Final Feature Set

Total Features After Engineering	58 (after dummy variable expansion)
Scaling	StandardScaler fitted on training data, applied to validation

8. MODEL BUILDING

8.1 Logistic Regression

Feature Selection — RFECV

Recursive Feature Elimination with Cross-Validation (RFECV) was applied using a Logistic Regression estimator with StratifiedKFold (n=4) and average_precision scoring. The optimal feature count was determined to be 11.

Selected features:

- number_of_vehicles_involved
- witnesses
- is_high_risk_hobby
- vehicle_to_injury_ratio
- policy_csl_250/500
- incident_type_Single Vehicle Collision
- collision_type_Side Collision
- incident_severity_Minor Damage
- incident_severity_Total Loss
- incident_severity_Trivial Damage
- property_damage_YES

Model was build using top 30 features in the first attempt, subsequently on the basis of p-value and vif some features - ['vehicle_claim', 'incident_type_Vehicle Theft', 'claim_to_premium_ratio', 'collision_type_Rear Collision', 'incident_city_Hillsdale'] were dropped

Model Summary (statsmodels GLM)

Multicollinearity Check	VIF max reduced to 3.83 (acceptable) after dropping 4 high-VIF features
--------------------------------	---

Key Significant Coefficients (p < 0.05)

Feature	Coefficient	Direction	Interpretation
is_high_risk_hobby	+2.14	→ Fraud	Strongest predictor — chess/cross-fit hobbies
incident_severity_Minor Damage	-1.90	→ Legit	Minor damage claims are less fraudulent
incident_severity_Total Loss	-1.68	→ Legit	Total loss claims are less fraudulent
incident_severity_Trivial Damage	-1.38	→ Legit	Trivial claims are less fraudulent
number_of_vehicles_involved	-0.53	→ Legit	If Lesser involved vehicles high likelihood of fraud
policy_csl_250/500	+0.47	→ Fraud	Higher coverage limit attracts fraud
insured_relationship_other-relative	+0.43	→ Fraud	If distant/other relative are involved then more likelihood of fraud

property_damage_YES	+0.40	→ Fraud	Property damage reported = fraud signal
vehicle_to_injury_ratio	+0.40	→ Fraud	Vehicle claim >> injury = staged accident
witnesses	+0.31	→ Fraud	More witnesses = organised staging

8.2 Random Forest (Tuned)

Base Model Observations

The initial Random Forest achieved 100% training accuracy — a clear sign of overfitting. Cross-validation (5-fold) accuracy of ~92% revealed the true generalisation ability. Feature importance scores were used to filter features (importance > 0.015 threshold).

Hyperparameter Tuning — GridSearchCV

n_estimators	Best: 70 (searched: 50, 60, 65, 70)
max_depth	Best: 22 (searched: 20, 22, 25)
min_samples_split	Best: 8 (searched: 8, 10)
min_samples_leaf	Best: 5 (searched: 5, 7)
max_leaf_nodes	Best: 70 (searched: 70, 75)
class_weight	Best: balanced_subsample
min_impurity_decrease	0.0005 [found by CV, but remove from CV params in notebook later]

9. MODEL EVALUATION & COMPARISON

9.1 Performance on Training Data

Metric	Logistic Regression	Random Forest (Tuned)
Accuracy	87.62%	93.43%
Sensitivity	88.57%	96.76%
Specificity	86.67%	90.10%
Precision	86.92%	90.71%
Recall	88.57%	96.76%
F1-Score	87.74%	93.64%
AUC-ROC	93.28%	98.91%

** Optimal cutoff for Logistic Regression was selected as 0.5 keeping in mind with 0.5 performance on validation(unseen data) was also good where as best outcomes for training data were observed at cutoff 0.45

9.2 Performance on Validation Data (Primary Benchmark)

Metric	Logistic Regression	Random Forest (Tuned)	Winner
Accuracy	83.67%	84.67%	RF (+1.0%)
Sensitivity	81.08%	81.08%	Tie
Specificity	84.51%	85.84%	RF (+1.3%)
Precision	63.16%	65.22%	RF (+2.1%)
Recall	81.08%	81.08%	Tie
F1-Score	71.01%	72.29%	RF (+1.3%)

AUC-ROC	0.877	0.893	RF (+1.6%)
----------------	-------	-------	------------

Recommended Model: Tuned Random Forest. It outperforms Logistic Regression on every validation metric. Its AUC-ROC of 0.893 means the model correctly ranks a random fraud claim above a legitimate claim ~89.3% of the time on unseen data.

9.3 Confusion Matrix Analysis — Validation Set

Random Forest (Tuned) — 299 validation records:

	Predicted: Legitimate	Predicted: Fraud
Actual: Legitimate	194 (True Negative)	32 (False Positive)
Actual: Fraud	14 (False Negative)	60 (True Positive)

- 60 of 74 actual fraud cases correctly identified (81.08% sensitivity).
- 32 legitimate claims flagged for review (false positive rate 14.2% — 1 in 7 legit claims).
- Only 14 fraud cases missed (false negatives) — the most costly error type.

9.4 ROC-AUC Interpretation

The ROC-AUC score measures the model's ability to discriminate between fraud and legitimate claims across all possible decision thresholds. Both models substantially outperform random guessing (AUC = 0.5):

LR Validation AUC	0.877 — strong discrimination; suitable for rule-extraction and explainability
RF Validation AUC	0.893 — best overall; recommended for production scoring
Training AUC (LR)	0.9328 — confirms strong learning; validation drop is normal
Random Baseline	0.500 — both models far exceed random guessing

10. EDA INSIGHTS CONFIRMED BY THE MODEL

A key strength of this analysis is that every major EDA observation found a corresponding statistical confirmation in the Logistic Regression model coefficients. This cross-validation between exploratory and confirmatory analysis builds high confidence in the findings.

EDA Finding	Model Confirmation	Coefficient / p-value
chess & cross-fit → ~80% fraud	Strongest predictor in LR	+2.14 (p<0.001)
Major Damage → more fraudulent	Minor/Trivial/Total Loss all negative	-1.38 to -1.90 (p<0.001)
vehicle_claim >> injury_claim in fraud	vehicle_to_injury_ratio significant	+0.40 (p<0.001)
Property damage YES = fraud signal	property_damage_YES significant	+0.40 (p<0.001)
More witnesses = organised staging	witnesses significant	+0.32 (p=0.002)
Higher policy limit = more fraud	policy_csl_250/500 significant	+0.47 (p<0.001)

Multi-vehicle = genuine accident	number_of_vehicles_involved negative	-0.52 (p=0.008)
---	---	-----------------

11. BUSINESS RECOMMENDATIONS

The following recommendations are structured across four operational areas and are directly grounded in the data and model findings. Each recommendation includes the supporting evidence and an indication of implementation priority.

11.1 Policy Underwriting

Fraud prevention begins at policy issuance. Several policyholder attributes identified as significant fraud predictors can be incorporated into underwriting scoring systems.

Recommendation U1 — Hobby Risk Scoring

Implement a hobby risk tier at application stage:

- Tier 1 (High Risk): chess, cross-fit → mandatory additional verification, extended waiting period for claims
- Tier 2 (Elevated): other hobbies with >35% fraud rate → flag for enhanced underwriting review
- Evidence: is_high_risk_hobby has a coefficient of +2.14 (p<0.001) — the single strongest predictor in the model.

Recommendation U2 — Education & Occupation Screening

- Apply enhanced due-diligence for applicants with Masters, PhD, or MD qualifications in executive-managerial.
- These segments combine the means, motive, and opportunity for sophisticated fraud schemes.

Recommendation U3 — Policy Coverage Limit Review

- CSL 250/500 policies show significantly elevated fraud rates (coeff +0.47, p<0.001).
- Introduce a mandatory claims-history review before issuing or renewing high-limit policies.
- Consider requiring additional identity verification for applicants seeking CSL 250/500 coverage.

Recommendation U4 — Early Claim Waiting Period

- Flag all claims filed within 60 days of policy binding.
- Consider a contractual 30-day waiting period for claims on new policies in high-risk segments.

11.2 Claims Processing

Claims-stage interventions can halt fraudulent payouts before they occur. The following automated rules can be derived directly from model findings.

Recommendation C1 — Automated Alert Rules

Implement the following real-time alert triggers at claim intake:

Vehicle-to-Injury Rule	Alert if vehicle_claim > 3× injury_claim — staged accident indicator (coeff +0.40)
Witness-Document Mismatch	Alert if witnesses ≥ 2 AND police_report_available = NO — organised staging (coeff +0.31)

Early Claim Rule	Mandatory review if days since policy binding < 60 days
No-Report Property Rule	Alert if property_damage = YES AND police_report_available = NO
High-Risk Hobby Rule	Auto-route to SIU if insured_hobbies ∈ {chess, cross-fit}

Recommendation C2 — Severity-Based Escalation

- Do not assume that moderate-severity (Major Damage) claims are less risky — the data shows the opposite.
- Escalate all Major Damage claims to Senior Adjusters for a secondary review.
- Introduce a standardised fraud checklist for Major Damage assessment that verifies physical evidence.

11.3 Geographic Operations

Fraud is not uniformly distributed. The data reveals clear geographic concentration that should drive resource allocation decisions.

Recommendation G1 — Law Enforcement Collaboration

- Partner with local police departments in OH and PA hotspot cities to establish fraud reporting channels.
- Share anonymised fraud patterns with authorities to help break up organised fraud rings.

Recommendation G2 — Fraud Awareness Campaigns

- Run targeted consumer fraud-awareness campaigns in high-risk regions to deter opportunistic fraud.
- Publicise prosecution outcomes in local media to increase perceived risk of committing fraud.

Recommendation G3 — Geographic Pricing Adjustment

- Review risk-based pricing models to ensure premiums in OH and PA reflect the elevated fraud exposure.
- Monitor fraud rates quarterly at city and zip-code level to detect emerging hotspot migration.

12. CONCLUSION

This analysis demonstrates that fraudulent insurance claims exhibit identifiable, statistically significant patterns that can be reliably detected using machine learning. The Tuned Random Forest model, achieving 84.67% accuracy and an AUC-ROC of 0.893 on the validation set, provides Global Insure with a production-ready tool to flag suspicious claims at the point of intake — before any payout is made.

The strongest fraud signals identified — hobby type (chess/cross-fit), incident severity paradox (Major Damage = more fraud), vehicle-to-injury claim ratio, geographic hotspots (OH & PA), and documentation gaps — are not only statistically confirmed by the model but are also operationally actionable without requiring complex infrastructure changes.

The recommendations in this report span the full insurance value chain: from underwriting risk scoring and claims automation rules, to geographic SIU deployment and long-term data enrichment strategy. Implemented together, they represent a comprehensive, evidence-based approach to reducing fraud losses and protecting honest policyholders.

Bottom Line: With the right model deployed and the right business rules in place, Global Insure can identify approximately 4 in 5 fraudulent claims before payment — transforming fraud detection from a reactive, post-payout investigation into a proactive, data-driven prevention capability.